
Statistical Inference

B. Statistical Data Science 2nd Year Indian Statistical Institute

Teacher: Soham Sarkar

Exercise Series 11

In the last class, we defined the family of distributions having *monotone likelihood ratio* (MLR) property. We also discussed about construction most powerful (MP) tests when the statistical model has MLR. In the following exercise, we will prove some of the results that we mentioned in the class.

Exercise 1. Recall that we say $\{p_\theta(\cdot) : \theta \in \Theta\}$ has monotone likelihood ratio (MLR) in $T(\cdot)$ if, for every $\theta_1, \theta_2 \in \Theta$ with $\theta_1 < \theta_2$, $R(\mathbf{x}) := \frac{p_{\theta_2}(\mathbf{x})}{p_{\theta_1}(\mathbf{x})}$ is monotone (non-increasing/non-decreasing) in $T(\mathbf{x})$.

That is, if $R(\mathbf{x}) = \frac{p_{\theta_2}(\mathbf{x})}{p_{\theta_1}(\mathbf{x})} =: g(T(\mathbf{x}))$, then $g(\cdot)$ is a monotone function.

- (a) Take any $\theta_a > \theta_0$. Suppose that $\frac{p_{\theta_a}(\mathbf{x})}{p_{\theta_0}(\mathbf{x})} =: g(T(\mathbf{x}))$ as defined above is non-decreasing. If $g(\cdot)$ is invertible, then show that $g^{-1}(\cdot)$ is non-decreasing.

Note: Notice that the function $g(\cdot)$ depends on θ_0 and θ_a . So, $g^{-1}(\cdot)$ also depends on θ_0 and θ_a .

- (b) Use part (a) to show that an MP level α test for $\mathcal{H}_0 : \theta = \theta_0$ vs. $\mathcal{H}_a : \theta = \theta_a$ ($\theta_a > \theta_0$) is given by

$$\varphi(\mathbf{x}) = \begin{cases} 1 & \text{if } T(\mathbf{x}) > c, \\ 0 & \text{if } T(\mathbf{x}) < c, \\ \gamma & \text{if } T(\mathbf{x}) = c, \end{cases} \quad \text{with } \mathbb{P}_{\theta_0}(T(\mathbf{X}) > c) + \gamma \mathbb{P}_{\theta_0}(T(\mathbf{X}) = c) = \alpha.$$

Hint: Start with the MP test construction using NP lemma. Connect with $T(\mathbf{x})$ using the MLR property.

When the function $g(\cdot)$ is not invertible, we define its *left inverse* as

$$g^-(k) = \inf\{T(\mathbf{x}) : R(\mathbf{x}) \geq k\}.$$

- (c) Show that the function $g^-(\cdot)$ defined above is non-decreasing. Moreover, for every k , there exists a c such that $g(t) > k \Leftrightarrow t > c$ and $g(t) < k \Leftrightarrow t < c$. Hence, deduce that even if the ratio $R(\mathbf{x}) := \frac{p_{\theta_a}(\mathbf{x})}{p_{\theta_0}(\mathbf{x})}$ is not invertible in $T(\mathbf{x})$, the test in part (b) is still an MP size α test for $\mathcal{H}_0 : \theta = \theta_0$ vs. $\mathcal{H}_a : \theta = \theta_a (> \theta_0)$.
- (d) Notice that the test in part (b) depends only on the statistic $T(\mathbf{X})$ and a constant c which satisfies the size condition. To satisfy the size condition, only the distribution of $T(\mathbf{X})$ under $\mathcal{H}_0$ is taken into account. Using this, deduce that the MP test in part (b) does not depend on the particular value of $\theta_a (> \theta_0)$. Therefore, the test in part (b) is uniformly most powerful (UMP) size α for testing $\mathcal{H}_0 : \theta = \theta_0$ vs. $\mathcal{H}_a : \theta > \theta_0$.
- (e) Recall that the MLR property ensures that the power of the test in part (b) is non-decreasing in θ . That is $\beta_\varphi(\theta) = \mathbb{E}_\theta \varphi(\mathbf{X})$ is non-decreasing in θ . Use this to deduce that the test in part (b) is UMP level α for testing $\mathcal{H}_0 : \theta \leq \theta_0$ vs. $\mathcal{H}_a : \theta > \theta_0$.

- (f) Derive the UMP level α test for testing $\mathcal{H}_0 : \theta \geq \theta_0$ vs. $\mathcal{H}_a : \theta < \theta_0$ when $\{p_\theta(\cdot) : \theta \in \Theta\}$ has MLR (increasing) in $T(\cdot)$.
- (g) Derive the UMP level α test for testing $\mathcal{H}_0 : \theta \leq \theta_0$ vs. $\mathcal{H}_a : \theta > \theta_0$ when $\{p_\theta(\cdot) : \theta \in \Theta\}$ has MLR (decreasing) in $T(\cdot)$.
- (h) Derive the UMP level α test for testing $\mathcal{H}_0 : \theta \geq \theta_0$ vs. $\mathcal{H}_a : \theta < \theta_0$ when $\{p_\theta(\cdot) : \theta \in \Theta\}$ has MLR (decreasing) in $T(\cdot)$.

Exercise 2. For the following examples, check whether the statistical model has MLR property. If so, then identify the statistic $T(\mathbf{X})$ w.r.t. which the model has MLR. Use this statistic to derive UMP test for testing (i) $\mathcal{H}_{0,\text{left}} : \theta \leq \theta_0$ vs. $\mathcal{H}_{a,\text{right}} : \theta > \theta_0$ and (ii) $\mathcal{H}_{0,\text{right}} : \theta \geq \theta_0$ vs. $\mathcal{H}_{a,\text{left}} : \theta < \theta_0$. Determine whether you need to randomize in order to achieve the desired level of significance α . In each case, draw the power function of the test. Comment on the utility of the test for testing $\mathcal{H}_0 : \theta = \theta_0$ vs. $\mathcal{H}_{a,\text{both}} : \theta \neq \theta_0$. Find UMP unbiased (UMPU) size α test for testing $\mathcal{H}_0 : \theta = \theta_0$ vs. $\mathcal{H}_{a,\text{both}} : \theta \neq \theta_0$.

- (a) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Binomial}(p), 0 \leq p \leq 1. \theta = p.$
- (b) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda), \lambda > 0. \theta = \lambda$
- (c) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Geometric}(p), 0 \leq p \leq 1. \theta = p.$
- (d) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}(0, \theta), \theta > 0.$
- (e) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(\mu, 1), \mu \in \mathbb{R}. \theta = \mu.$
- (f) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(0, \sigma^2), \sigma > 0.$
- (g) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Exponential}(\theta), \theta > 0.$
- (h) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Laplace}(\mu, 1), \mu \in \mathbb{R}. \theta = \mu.$
- (i) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Laplace}(0, \sigma), \sigma > 0. \theta = \sigma.$
- (j) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(\theta, \theta^2), \theta \in \mathbb{R}.$
- (k) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(\theta, \theta), \theta > 0.$